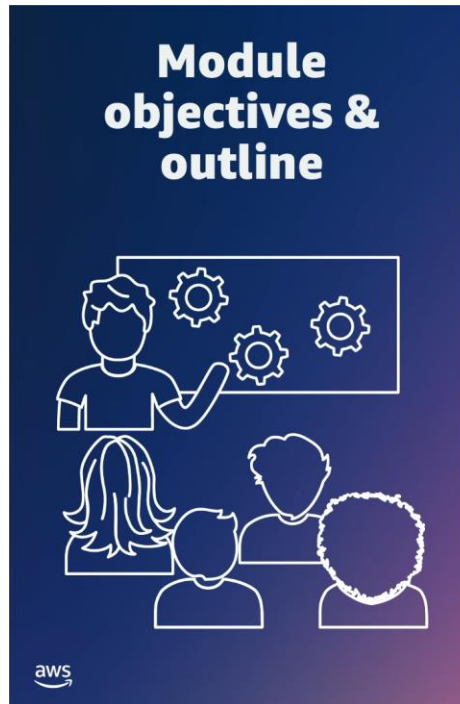


5.5 Introduction to Information Retrieval with Amazon Q



On completion, you will be able to:

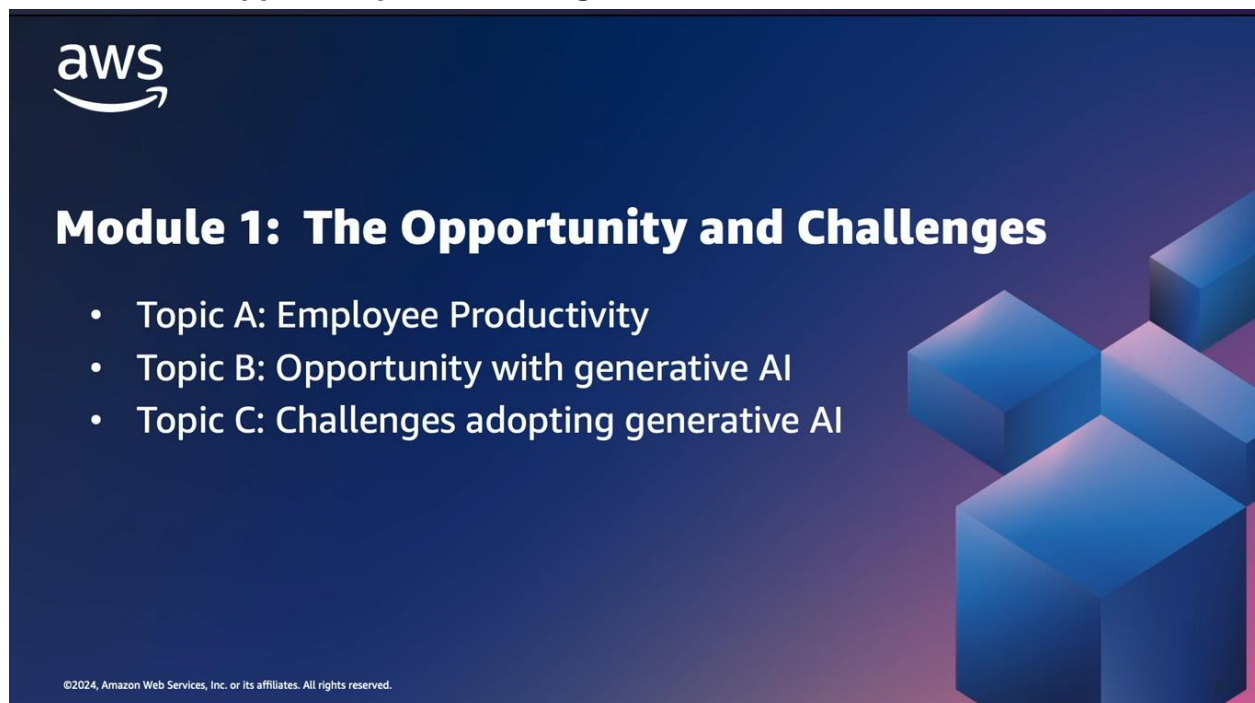
- Be able to why upskilling in generative AI increases productivity.
- Understand the different versions of Amazon Q and the appropriate use cases.
- Gain a practical understanding of how to use Amazon Q for AWS Builder Use improve developer productivity and creativity.

Topics:

- Module 1 The Opportunity and Challenges
- Module 2 What is Amazon Q?
- Module 3 Conclusion

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Module 1: The Opportunity and Challenges



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Employee productivity

Marketing Manager

Sales Manager

R&D Engineer

Financial Analyst

Developer

Help Desk Employee (IT, HR etc.)

Product Manager...



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Opportunity with generative AI



Generative AI's ability to understand natural language enables automation for work activities that account for

25%



Imagine a generative AI powered assistant that saves you

2 hours every day

[McKinsey Digital June 2023](#)



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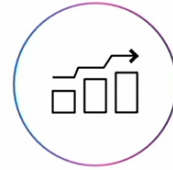
Challenges adopting generative AI



Accuracy



Security & control



Time to value



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Amazon Q addresses generative AI challenges



Accuracy

Accurate answers from
enterprise data sources
using RAG
Citations for fact-checking
Broad set of capabilities



Security & control

Support for enterprise
permissions
Ability to configure
response guardrails



Time to value

Connectors to 40+ popular
data sources
In-built vector index with
managed ingestion
In-built application
with SSO



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Module 2.1: What is Amazon Q?



Module 2: What is Amazon Q?

- Topic A: Generative AI Stack
- Topic B: Amazon Q Areas of Expertise
- Topic C: Amazon Q for Business Use
- Topic D: Amazon Q for AWS Builder Use

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Generative AI Stack

APPLICATIONS THAT LEVERAGE LLMs AND OTHER FMs



TOOLS TO BUILD WITH LLMs AND OTHER FMs



INFRASTRUCTURE FOR FM TRAINING AND INFERENCE



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Module 2.2: Introducing Amazon Q

Introducing

Amazon Q

**Your generative AI assistant designed
for work that can be tailored to your
business, data, code, and operations**



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Amazon Q areas of expertise

**Your
business**



**Building
on AWS**



**Amazon
QuickSight**



**Amazon
Connect**



**AWS
Supply Chain**



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Key features (for Business Use)

End Users



Conversational question-answering on enterprise data



Upload files and analyze content



Execute actions across enterprise apps



Generate content

Admins



Fully managed solution



Pre-built connectors, vector index and end user application



Permissions-aware responses



Customize and control Amazon Q with guardrails



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Key features – Conversational Q&A

LET'S GET TO WORK



Trusted answers generated from enterprise data



In-context conversations



Source references for fact-checking



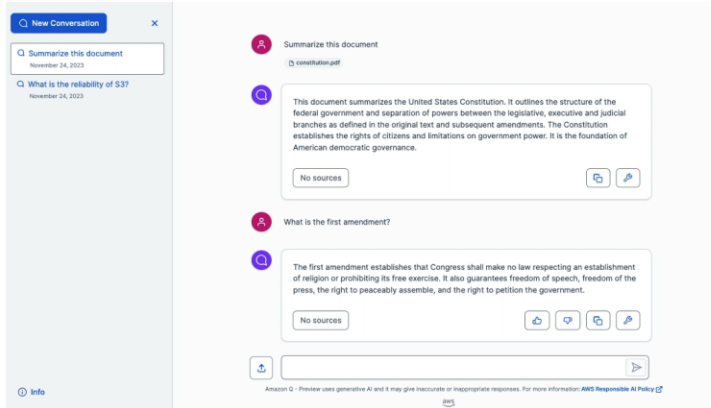
Conversation history

The screenshot displays the Amazon Q conversational Q&A interface. On the left, a sidebar titled "New Conversation" shows a list of previous questions, including "What is the reliability of S3?". The main area shows a conversation thread. The user asks, "What is the reliability of S3?". The assistant responds with a detailed answer about S3's 99.999999999% durability and 11 nines availability, mentioning its multi-region, multi-availability zone architecture and 11 nines availability. The user then asks, "Tell me more about availability zones". The assistant explains that availability zones are independent infrastructure units within a region, designed for high availability and fault tolerance, and are connected via low-latency links. The user asks, "How many availability zones should my data be stored across?". The assistant recommends storing data across multiple availability zones to achieve higher availability and fault tolerance, listing key reasons: availability zones are isolated locations, using multiple zones helps maximize availability, and critical data should be stored in multiple zones. The interface includes a "Sources" button and a "Feedback" button at the bottom of each response.



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Key features - Upload files and analyze content

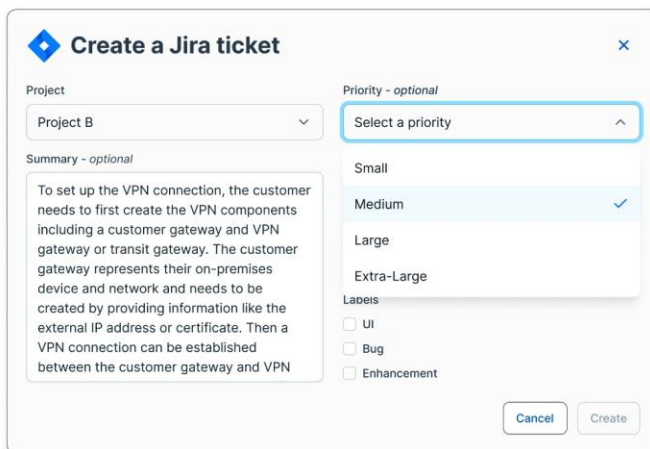


- Summarize documents
- Ask questions about content in the documents
- Analyze data
- Multiple file types supported – pdf, docx, csv, ppt, txt and more



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Key features - Execute actions using plugins



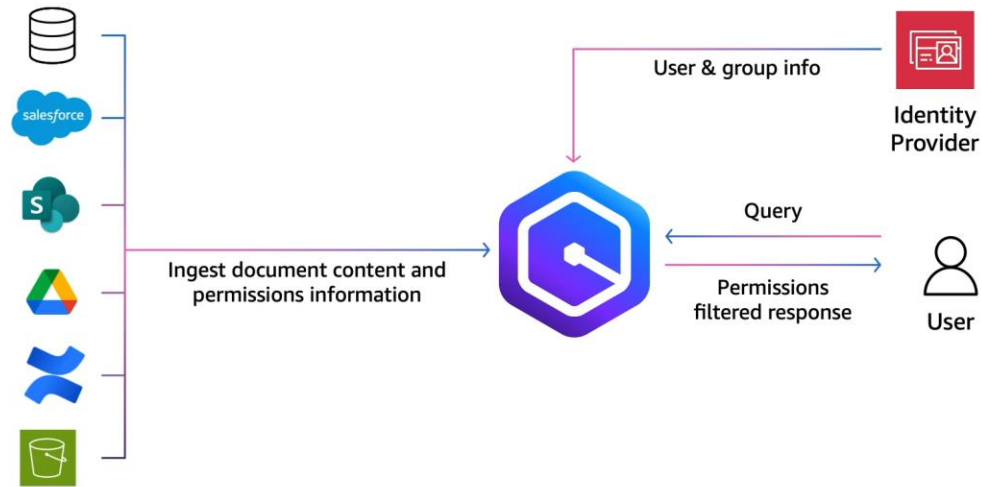
- Enable end-users to perform actions on SaaS applications



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Key features – Safety and security

BUSINESS Q IS AWARE OF ENTERPRISE USER PERMISSIONS



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Module 2.3: Amazon Q areas of expertise

Module 3: Conclusion